

6.2 The subspaces we need

This section gathers the vector-space facts used in the rest of the chapter. Readers who have worked through [Chapter 1](#) can skim the definitions. The results on orthogonal complements, especially [Lemma 6.2.1](#) and [Theorem 6.2.3](#), are used constantly afterwards and are proved in full.

6.2.1 Subspaces, spans and dimension

A nonempty set $\mathcal{S} \subseteq \mathbb{R}^n$ is a **subspace** if it is closed under addition and scalar multiplication: $\mathbf{u}, \mathbf{v} \in \mathcal{S}$ and $a, b \in \mathbb{R}$ imply $a\mathbf{u} + b\mathbf{v} \in \mathcal{S}$. Every subspace contains $\mathbf{0}$. The **span** of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ is the set of all their linear combinations. It is the smallest subspace containing them. A set of vectors is *linearly independent* if the only linear combination equal to $\mathbf{0}$ is the one with all coefficients zero. A **basis** for $\mathcal{S}$ is a linearly independent set that spans $\mathcal{S}$. All bases of $\mathcal{S}$ have the same number of elements, the **dimension** $\dim \mathcal{S}$.

For an $n \times p$ matrix $\mathbf{A}$ there are two subspaces we use constantly:

$$\begin{aligned}\mathcal{C}(\mathbf{A}) &= \{\mathbf{A}\mathbf{b} : \mathbf{b} \in \mathbb{R}^p\} \subseteq \mathbb{R}^n && \text{(column space),} \\ \mathcal{N}(\mathbf{A}) &= \{\mathbf{b} \in \mathbb{R}^p : \mathbf{A}\mathbf{b} = \mathbf{0}\} \subseteq \mathbb{R}^p && \text{(null space).}\end{aligned}$$

The **rank** of $\mathbf{A}$ is $\text{rank}(\mathbf{A}) = \dim \mathcal{C}(\mathbf{A})$. Two standard facts from [Chapter 1](#): row rank equals column rank, so $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^\top)$; and the rank–nullity theorem

$$\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A}) = p. \quad (6.2.1)$$

In regression, $\mathcal{C}(\mathbf{X})$ is the **model space**. $\mathcal{N}(\mathbf{X})$ is the set of directions in which $\boldsymbol{\beta}$ can move without changing $\mathbf{X}\boldsymbol{\beta}$. It is $\{\mathbf{0}\}$ exactly when $\mathbf{X}$ has full column rank.

Example. A rank-deficient layout

Suppose n observations fall into g groups and $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g]$, where $\mathbf{z}_k$ is the indicator of group k . Since $\mathbf{z}_1 + \dots + \mathbf{z}_g = \mathbf{1}$, the vector $(1, -1, \dots, -1)^\top$ lies in $\mathcal{N}(\mathbf{X})$. The g indicators are linearly independent (their supports are disjoint), so $\text{rank}(\mathbf{X}) = g$ while $p = g + 1$. By [Eq. 6.2.1](#), $\mathcal{N}(\mathbf{X})$ is exactly the line spanned by $(1, -1, \dots, -1)^\top$. The model space $\mathcal{C}(\mathbf{X}) = \mathcal{C}([\mathbf{z}_1, \dots, \mathbf{z}_g])$ is the set of vectors that are constant within groups.

6.2.2 Inner products, orthogonality and complements

Unless stated otherwise $\mathbb{R}^n$ carries the Euclidean inner product $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v}$ and norm $\|\mathbf{u}\| = (\mathbf{u}^\top \mathbf{u})^{1/2}$. Vectors are **orthogonal**, written $\mathbf{u} \perp \mathbf{v}$, if $\mathbf{u}^\top \mathbf{v} = 0$. The Cauchy–Schwarz inequality $|\mathbf{u}^\top \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$, with equality iff one vector is a multiple of the other, lets us define the angle $\theta \in [0, \pi]$ between nonzero vectors by

$$\cos \theta = \frac{\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}. \quad (6.2.2)$$

The angle comes back in [Section 6.7](#), where R^2 turns out to be a squared cosine. The Pythagorean identity

$$\mathbf{u} \perp \mathbf{v} \implies \|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \quad (6.2.3)$$

follows by expanding $(\mathbf{u} + \mathbf{v})^\top (\mathbf{u} + \mathbf{v})$. The cross terms are the ones that vanish.

Definition 6.2.1. Orthogonal complement

The **orthogonal complement** of a subspace $\mathcal{S} \subseteq \mathbb{R}^n$ is

$$\mathcal{S}^\perp = \{\mathbf{v} \in \mathbb{R}^n : \mathbf{v}^\top \mathbf{u} = 0 \text{ for all } \mathbf{u} \in \mathcal{S}\}.$$

It is immediate that $\mathcal{S}^\perp$ is a subspace and that $\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$: a vector orthogonal to itself has zero length. To test whether $\mathbf{v} \in \mathcal{S}^\perp$ it is enough to check orthogonality to a spanning set of $\mathcal{S}$. In particular, $\mathbf{v} \in \mathcal{C}(\mathbf{A})^\perp$ iff $\mathbf{A}^\top \mathbf{v} = \mathbf{0}$.

Lemma 6.2.1

For any $n \times p$ matrix $\mathbf{A}$,

$$\mathcal{N}(\mathbf{A}^\top) = \mathcal{C}(\mathbf{A})^\perp \quad \text{and} \quad \mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)^\perp.$$

Proof. $\mathbf{v} \in \mathcal{N}(\mathbf{A}^\top)$ means $\mathbf{A}^\top \mathbf{v} = \mathbf{0}$, that is, $\mathbf{v}$ is orthogonal to every column of $\mathbf{A}$, which is the same as $\mathbf{v} \perp \mathcal{C}(\mathbf{A})$. The second identity is the first applied to $\mathbf{A}^\top$. ■

6.2.3 Orthonormal bases and Gram–Schmidt

Vectors $\mathbf{u}_1, \dots, \mathbf{u}_k$ are **orthonormal** if $\mathbf{u}_i^\top \mathbf{u}_j$ is 1 for $i = j$ and 0 otherwise. Orthonormal vectors are linearly independent: taking the inner product of $\sum_j a_j \mathbf{u}_j = \mathbf{0}$ with $\mathbf{u}_i$ gives $a_i = 0$. The coordinates of a vector $\mathbf{v} \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ are simply the inner products $a_i = \mathbf{u}_i^\top \mathbf{v}$, with no linear system to solve. This is why orthonormal bases are so convenient.

Proposition 6.2.2. Gram–Schmidt

Let $\mathbf{v}_1, \dots, \mathbf{v}_k$ be linearly independent. Define recursively

$$\mathbf{w}_j = \mathbf{v}_j - \sum_{i < j} (\mathbf{u}_i^\top \mathbf{v}_j) \mathbf{u}_i, \quad \mathbf{u}_j = \mathbf{w}_j / \|\mathbf{w}_j\|, \quad j = 1, \dots, k.$$

Then each $\mathbf{w}_j \neq \mathbf{0}$. The vectors $\mathbf{u}_1, \dots, \mathbf{u}_k$ are orthonormal, and $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_j) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ for every j .

Proof. Induction on j . Suppose the claims hold for $j - 1$. The vector $\mathbf{w}_j$ is $\mathbf{v}_j$ minus an element of $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$. If $\mathbf{w}_j$ were zero, $\mathbf{v}_j$ would lie in that span, contradicting independence. For $l < j$, $\mathbf{u}_l^\top \mathbf{w}_j = \mathbf{u}_l^\top \mathbf{v}_j - \sum_{i < j} (\mathbf{u}_i^\top \mathbf{v}_j) \mathbf{u}_l^\top \mathbf{u}_i = \mathbf{u}_l^\top \mathbf{v}_j - \mathbf{u}_l^\top \mathbf{v}_j = 0$, so $\mathbf{u}_j$ is orthogonal to its predecessors. Finally, $\mathbf{u}_j \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}, \mathbf{v}_j) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ and $\mathbf{v}_j \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_j)$, so the two spans agree.

Two consequences are used repeatedly. Every subspace has an orthonormal basis (apply Gram–Schmidt to any basis). And any orthonormal basis of a subspace $\mathcal{S}$ can be extended to an orthonormal basis of $\mathbb{R}^n$: extend it to a basis of $\mathbb{R}^n$ in any way, then apply Gram–Schmidt, which leaves the first vectors unchanged. [Section 6.10](#) shows that Gram–Schmidt, written in matrix form, is the QR factorization. ■

6.2.4 Direct sums and the orthogonal decomposition

For subspaces $\mathcal{S}, \mathcal{T}$, the sum $\mathcal{S} + \mathcal{T} = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in \mathcal{S}, \mathbf{v} \in \mathcal{T}\}$ is a subspace. When $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$ the sum is *direct*, written $\mathcal{S} \oplus \mathcal{T}$. Every element then decomposes as $\mathbf{u} + \mathbf{v}$ in exactly one way: if $\mathbf{u} + \mathbf{v} = \mathbf{u}' + \mathbf{v}'$ then $\mathbf{u} - \mathbf{u}' = \mathbf{v}' - \mathbf{v}$ lies in both subspaces and so is zero. When in addition $\mathcal{S} \perp \mathcal{T}$, the sum is an *orthogonal* direct sum.

Theorem 6.2.3. Orthogonal decomposition

For every subspace $\mathcal{S} \subseteq \mathbb{R}^n$,

$$\mathbb{R}^n = \mathcal{S} \oplus \mathcal{S}^\perp, \quad \dim \mathcal{S} + \dim \mathcal{S}^\perp = n, \quad (\mathcal{S}^\perp)^\perp = \mathcal{S}.$$

Hence every $\mathbf{y} \in \mathbb{R}^n$ can be written uniquely as $\mathbf{y} = \mathbf{u} + \mathbf{v}$ with $\mathbf{u} \in \mathcal{S}$ and $\mathbf{v} \perp \mathcal{S}$.

Proof. Let $\mathbf{u}_1, \dots, \mathbf{u}_r$ be an orthonormal basis of $\mathcal{S}$, extended to an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_n$ of $\mathbb{R}^n$. A vector $\mathbf{y} = \sum_{i=1}^n a_i \mathbf{u}_i$ is orthogonal to $\mathcal{S}$ iff $a_1 = \dots = a_r = 0$, so $\mathcal{S}^\perp = \text{span}(\mathbf{u}_{r+1}, \dots, \mathbf{u}_n)$ has dimension $n - r$. Writing $\mathbf{y} = \sum_{i \leq r} a_i \mathbf{u}_i + \sum_{i > r} a_i \mathbf{u}_i$ exhibits the decomposition, and it is unique because $\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$. By the same argument with the roles of the two index sets swapped, $(\mathcal{S}^\perp)^\perp = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_r) = \mathcal{S}$. ■

Combining [Theorem 6.2.3](#) with [Lemma 6.2.1](#) gives a matrix fact used in almost every later chapter.

Corollary 6.2.4

For any $n \times p$ matrix $\mathbf{A}$:

- (a) $\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A})$;
- (b) $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)$ and $\mathcal{C}(\mathbf{A} \mathbf{A}^\top) = \mathcal{C}(\mathbf{A})$;
- (c) $\text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{A} \mathbf{A}^\top) = \text{rank}(\mathbf{A})$;
- (d) $\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top)$ and $\mathbb{R}^p = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$.

Proof.

- (a) If $\mathbf{A} \mathbf{b} = \mathbf{0}$ then $\mathbf{A}^\top \mathbf{A} \mathbf{b} = \mathbf{0}$. Conversely, $\mathbf{A}^\top \mathbf{A} \mathbf{b} = \mathbf{0}$ gives $\mathbf{b}^\top \mathbf{A}^\top \mathbf{A} \mathbf{b} = \|\mathbf{A} \mathbf{b}\|^2 = 0$, so $\mathbf{A} \mathbf{b} = \mathbf{0}$.

- (b) By [Lemma 6.2.1](#) applied to the symmetric matrix $\mathbf{A}^\top \mathbf{A}$, and then (a) and [Lemma 6.2.1](#) again, $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A}^\top \mathbf{A})^\perp = \mathcal{N}(\mathbf{A})^\perp = (\mathcal{C}(\mathbf{A}^\top)^\perp)^\perp = \mathcal{C}(\mathbf{A}^\top)$, using $(\mathcal{S}^\perp)^\perp = \mathcal{S}$ in the last step. The second identity is the first applied to $\mathbf{A}^\top$.
- (c) Take dimensions in (b) and use $\text{rank}(\mathbf{A}^\top) = \text{rank}(\mathbf{A})$.
- (d) This is [Lemma 6.2.1](#) and [Theorem 6.2.3](#). ■

Part (b) is the reason the normal equations $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$ can always be solved: the right-hand side lies in $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$. We come back to this in [Section 6.4](#).

Lemma 6.2.5

For subspaces $\mathcal{S}_1, \mathcal{S}_2 \subseteq \mathbb{R}^n$,

$$(\mathcal{S}_1 + \mathcal{S}_2)^\perp = \mathcal{S}_1^\perp \cap \mathcal{S}_2^\perp \quad \text{and} \quad (\mathcal{S}_1 \cap \mathcal{S}_2)^\perp = \mathcal{S}_1^\perp + \mathcal{S}_2^\perp.$$

Proof. A vector is orthogonal to $\mathcal{S}_1 + \mathcal{S}_2$ iff it is orthogonal to every $\mathbf{u}_1 + \mathbf{u}_2$, which (taking $\mathbf{u}_2 = \mathbf{0}$ or $\mathbf{u}_1 = \mathbf{0}$) is the case iff it is orthogonal to both subspaces. For the second identity, apply the first to $\mathcal{S}_1^\perp$ and $\mathcal{S}_2^\perp$: $(\mathcal{S}_1^\perp + \mathcal{S}_2^\perp)^\perp = \mathcal{S}_1 \cap \mathcal{S}_2$, and take complements of both sides. ■

[Lemma 6.2.5](#) is what we need to describe reduced models defined by linear constraints ([Theorem 6.5.2](#)).

6.2.5 Exercises

6.2.5.1 A. Check your understanding

Exercise. A1

Show that $\mathcal{S}^\perp$ is a subspace for any set $\mathcal{S}$ (not necessarily a subspace), and that $\mathcal{S} \subseteq \mathcal{T}$ implies $\mathcal{T}^\perp \subseteq \mathcal{S}^\perp$. Show that $\mathcal{S}^\perp = \text{span}(\mathcal{S})^\perp$.

6.2.5.2 B. Practice

Exercise. B1

A one-way layout has three groups of sizes 2, 2, 1 and model matrix $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3]$. Find bases for $\mathcal{C}(\mathbf{X})$, $\mathcal{N}(\mathbf{X})$, $\mathcal{C}(\mathbf{X}^\top)$ and $\mathcal{C}(\mathbf{X})^\perp$, and check that the dimensions agree with [Eq. 6.2.1](#) and [Theorem 6.2.3](#).

Exercise. B2

Show that $\mathcal{C}(\mathbf{AB}) \subseteq \mathcal{C}(\mathbf{A})$, and deduce $\text{rank}(\mathbf{AB}) \leq \min\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\}$. Show that $\mathcal{C}(\mathbf{AB}) = \mathcal{C}(\mathbf{A})$ whenever $\mathbf{B}$ has full row rank.

6.2.5.3 C. Going deeper**Exercise. C1**

For subspaces $\mathcal{S}, \mathcal{T} \subseteq \mathbb{R}^n$, prove $\dim(\mathcal{S} + \mathcal{T}) = \dim \mathcal{S} + \dim \mathcal{T} - \dim(\mathcal{S} \cap \mathcal{T})$. Use it to show that two r -dimensional subspaces of $\mathbb{R}^n$ with $2r > n$ must share a nonzero vector.